Exact Orthogonal-CI Contraction for Arbitrary Selected Valence-Bond Structure Spaces

Theory and implementation note

16 October 2026

Abstract

An arbitrary selected valence-bond (VB) structure space is generally not invariant under orbital orthogonalization. This does not prevent exact evaluation of its Hamiltonian and overlap actions through an orthogonal configuration-interaction (CI) representation. The transformed wave function must be retained, but its complete spin-string product array need not be formed. We derive an exact source-column factorization, apply the Slater–Condon one-body generators separately to the two spin factors, and contract only the original products needed for the final structure-space projection. The resulting equations preserve the selected variational space without a closure assumption or intermediate amplitude truncation. Source and channel tiling bound the numerical workspace. Optional spin-exchange reduction and causal pruning of the exterior adjoint are derived separately. The scope is the Hamiltonian/overlap action at a fixed orbital point.

1 The selected VB space

Let n be the number of active spatial orbitals, with n_α and n_β active electrons of each spin. The complete one-spin string spaces have dimensions

$$U_\alpha = \binom{n}{n_\alpha}, \qquad U_\beta = \binom{n}{n_\beta}. \tag{1}$$

An index a identifies an α string and b a β string. For a selected VB structure I, let $w_{ab,I}$ be its signed expansion coefficient on the corresponding spin-string product, with all normalization conventions included. For structure coefficients $\mathbf{C}$, define

$$X = \mathcal{W}\mathbf{C}, \qquad X_{ab} = \sum_I w_{ab,I} C_I. \tag{2}$$

Here $\mathcal{W}$ is a sparse linear map from structure coefficients to a matrix of spin-string product coefficients. The matrix notation does not prescribe a dense allocation: the map can be stored as sparse incidences between structures and touched string products. It introduces no determinant-pair traversal or separately optimized determinant coefficient vector.

The original support and its two marginal index sets are

$$\Omega = \{(a,b) : w_{ab,I} \neq 0 \text{ for at least one selected } I\}, \qquad t = |\Omega|, \tag{3}$$

$$A_\Omega = \{a : \exists b,\ (a,b) \in \Omega\}, \qquad J = \{b : \exists a,\ (a,b) \in \Omega\}, \qquad r = |J|. \tag{4}$$

Support is determined by the selected structures, not by numerical smallness or cancellations in a particular Davidson vector. Using the Frobenius inner product, the adjoint structure map is

$$(\mathcal{W}^* Z)_I = \sum_{(a,b)\in\Omega} w_{ab,I} Z_{ab}. \tag{5}$$

Thus only entries in Ω are needed at the *end* of a complete action. Nothing in this statement permits dropping intermediate entries outside Ω.

2 Orthogonalization without a closure assumption

At a fixed orbital point, write the nonorthogonal active orbitals as $\phi = \chi R$, where χ is an orthonormal basis of the same active one-particle space. For example, if O is their positive-definite overlap,

$$O = R^{\mathsf{T}} R, \qquad \chi = \phi R^{-1}. \tag{6}$$

This is an evaluation-coordinate change, not a change to the physical localization constraints or to the orbital optimization variables. Multilinearity of antisymmetrized products gives the exterior maps

$$T_\sigma = \wedge^{n_\sigma} R, \qquad (T_\sigma)_{ai} = \det R[a, i], \qquad \sigma \in \{\alpha, \beta\}, \tag{7}$$

where the row and column subsets contain n_σ orbitals with a fixed ordering convention. The orthogonal-basis coefficient matrix is

$$Y = \mathcal{T} X = T_\alpha X T_\beta^{\mathsf{T}}, \qquad \mathcal{T}^* Z = T_\alpha^{\mathsf{T}} Z T_\beta. \tag{8}$$

The minor matrices T_σ are mathematical definitions; the implementation applies elementary exterior transforms without assembling them densely.

Let $\mathcal{H}$ be the active Hamiltonian action in the orthogonal spin-string product basis. The selected structure Hamiltonian and overlap are exactly

$$\boxed{H = \mathcal{W}^* \mathcal{T}^* \mathcal{H} \mathcal{T} \mathcal{W}, \qquad M = \mathcal{W}^* \mathcal{T}^* \mathcal{T} \mathcal{W}.} \tag{9}$$

An orbital-dependent scalar inactive-plus-nuclear energy E_0 contributes $E_0 M$ to the total structure Hamiltonian and is separate from the active action H in this note. The variational energy is

$$E = E_0 + \frac{\mathbf{C}^{\mathsf{T}} H \mathbf{C}}{\mathbf{C}^{\mathsf{T}} M \mathbf{C}}, \tag{10}$$

with H denoting the active contribution of Eq. (9).

Proposition 1 (Exact selected-space restriction)**.** *For any selected structure map $\mathcal{W}$, Eq.* (9) *gives the exact selected-space Hamiltonian and overlap, whether or not $\mathcal{T}$* $\operatorname{ran} \mathcal{W} \subseteq \operatorname{ran} \mathcal{W}$.

Proof. For arbitrary structure vectors $\mathbf{u}$ and $\mathbf{v}$, the two physical states have orthogonal-basis amplitudes $\mathcal{T}\mathcal{W}\mathbf{u}$ and $\mathcal{T}\mathcal{W}\mathbf{v}$. Their Hamiltonian matrix element is

$$\langle \mathcal{T}\mathcal{W}\mathbf{u}, \mathcal{H}(\mathcal{T}\mathcal{W}\mathbf{v}) \rangle_{\mathrm{F}} = \mathbf{u}^{\mathsf{T}} \big(\mathcal{W}^* \mathcal{T}^* \mathcal{H} \mathcal{T} \mathcal{W} \big) \mathbf{v}. \tag{11}$$

Replacing $\mathcal{H}$ by the identity gives the overlap. No intermediate restriction to $\operatorname{ran} \mathcal{W}$ was inserted. □

The nonclosure issue arises only if a different operation is used, for example by zeroing transformed amplitudes outside the original selected support before applying $\mathcal{H}$. That operation changes the wave function. Equation (9) does not do so. Davidson vectors and Ritz coefficients remain in the selected VB structure space; the orthogonal representation is used internally to evaluate their images. For linearly independent selected structures, their eigenproblem remains

$$H\mathbf{C} = E_{\mathrm{act}} M \mathbf{C}. \tag{12}$$

The adjoint $\mathcal{W}^*$ contracts against the selected test structures; it is not an inverse coefficient transform. The nonidentity overlap M is retained, so this procedure does not replace the selected VB problem by an unconstrained orthogonal-CI eigenproblem.

2.1 A minimal nonclosed example

Consider a two-by-two coefficient matrix with the single selected product $X = e_2 e_2^{\mathsf{T}}$, and let

$$T_\alpha = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}, \qquad T_\beta = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}. \tag{13}$$

This is a linear-algebra example of the coefficient map, not an assertion that a single spin product is a spin-adapted VB structure. Its transformed amplitudes are

$$Y = \begin{pmatrix} ab & a \\ b & 1 \end{pmatrix} = \begin{pmatrix} a \\ 1 \end{pmatrix} \begin{pmatrix} b \\ 1 \end{pmatrix}^{\mathsf{T}}. \tag{14}$$

For nonzero a and b, all four products are present, so the original one-dimensional selected space is not closed. Nevertheless, two vectors represent the complete Y exactly. The overlap is $(1 + a^2)(1 + b^2)$, not the value 1 obtained by retaining only Y_{22}. An exact Hamiltonian contraction must likewise use all transformed amplitudes, explicitly or implicitly. The factorized route below retains them implicitly.

3 Exact source factors

Order the original touched β columns as $J = (j_1, \ldots, j_r)$, and set

$$A_{:\ell} = X_{:j_\ell}, \qquad B_{:\ell} = e_{j_\ell}, \qquad X = AB^{\mathsf{T}}. \tag{15}$$

This is an identity for every input vector. It is not a truncated singular value decomposition, and r is not a fitted rank. Transforming only the two factors gives

$$L = T_\alpha A, \qquad F = T_\beta B, \qquad \mathcal{T}X = LF^{\mathsf{T}}. \tag{16}$$

The factors have dimensions $U_\alpha \times r$ and $U_\beta \times r$. The complete transformed $U_\alpha \times U_\beta$ matrix need not be formed, even if every one of its entries is nonzero.

The distinction between r and the number of structures matters. A single covalent structure with m disjoint singlet pairs generally touches 2^m original source columns. Several structures can share most of those columns, and a broad selected space can touch all U_β columns. Thus rank $X \le r$ is an upper bound, not an identification of r with one, with rank X, or with the number of structures.

4 Hamiltonian action on the factors

4.1 One-body channels and the complete active Hamiltonian

Use real spatial orbitals and real, spin-independent molecular integrals. Let $e_{pq} = \sum_\sigma a^\dagger_{p\sigma} a_{q\sigma}$, and use the chemists' notation $(pq|rs)$ for two-electron integrals. The active Hamiltonian is

$$\hat{H} = \sum_{pq} h_{pq} e_{pq} + \frac{1}{2} \sum_{pqrs} (pq|rs)\big(e_{pq} e_{rs} - \delta_{qr} e_{ps}\big). \tag{17}$$

Here h includes the effective one-electron contribution of the inactive orbitals. Define

$$f_{ps} = h_{ps} - \frac{1}{2} \sum_q (pq|qs), \qquad N = n_\alpha + n_\beta > 0. \tag{18}$$

Packed pair labels $P = (p, q)$ have $p \ge q$, so there are $d = n(n+1)/2$ channels. Define the symmetric generators

$$\hat{E}_{\sigma P} = \begin{cases} a^\dagger_{p\sigma} a_{p\sigma}, & p = q, \\ a^\dagger_{p\sigma} a_{q\sigma} + a^\dagger_{q\sigma} a_{p\sigma}, & p > q, \end{cases} \qquad \hat{E}_P = \hat{E}_{\alpha P} + \hat{E}_{\beta P}. \tag{19}$$

Let $f_P = f_{pq}$, $V_{PQ} = (pq|rs)$ for $Q = (r, s)$, and $d_P = 1$ for diagonal pairs and zero otherwise. A symmetric kernel containing the *entire* active Hamiltonian is

$$K_{PQ} = \frac{1}{2}V_{PQ} + \frac{f_P d_Q + d_P f_Q}{2N}, \qquad \hat{H} = \sum_{PQ} K_{PQ}\hat{E}_P\hat{E}_Q. \tag{20}$$

Indeed, $\sum_P d_P\hat{E}_P = \hat{N} = N\hat{I}$ on the fixed-electron space. The second term in K therefore gives $\sum_P f_P\hat{E}_P$; the first term supplies the quadratic part of Eq. (17). The usual integral permutation symmetries account for both ordered off-diagonal orbital pairs through $\hat{E}_P$; no extra off-diagonal multiplicity is needed in K.

Let $E_{\sigma P}$ be the matrix of $\hat{E}_{\sigma P}$ on its one-spin string space. The Slater–Condon rules make this operation a sparse signed occupation/replacement scatter, rather than a dense string pair contraction. Its action on a product matrix is

$$\mathcal{E}_P(Z) = E_{\alpha P}Z + ZE_{\beta P}^{\mathsf{T}}, \qquad \mathcal{H} = \sum_{PQ} K_{PQ}\mathcal{E}_P\mathcal{E}_Q. \tag{21}$$

These identities require a complete carrier for each fixed-spin sector, or an exactly equivalent representation of all required intermediates. Arbitrary *selected VB structure spaces* need not be complete. The two statements are different.

4.2 Separation of the two spin factors

Define the same-spin channel actions

$$D_\sigma = \sum_{PQ} K_{PQ}E_{\sigma P}E_{\sigma Q}. \tag{22}$$

They include the particular distribution of one-electron terms implied by Eq. (20); they are not independently assigned physical α- and β-spin Hamiltonians. Expanding the two generators gives

$$\begin{aligned}\mathcal{E}_P\mathcal{E}_Q(LF^{\mathsf{T}}) = {} & (E_{\alpha P}E_{\alpha Q}L)F^{\mathsf{T}} + L(E_{\beta P}E_{\beta Q}F)^{\mathsf{T}} \\ & + (E_{\alpha P}L)(E_{\beta Q}F)^{\mathsf{T}} + (E_{\alpha Q}L)(E_{\beta P}F)^{\mathsf{T}}.\end{aligned} \tag{23}$$

The last two terms become identical after summation because $K_{PQ} = K_{QP}$. Consequently,

$$\boxed{\mathcal{H}(LF^{\mathsf{T}}) = (D_\alpha L)F^{\mathsf{T}} + L(D_\beta F)^{\mathsf{T}} + 2\sum_{PQ} K_{PQ}(E_{\alpha P}L)(E_{\beta Q}F)^{\mathsf{T}}.} \tag{24}$$

The proof uses commutativity of different spin sectors, not commutativity of generators within one spin sector. It applies to unequal U_α, U_β, unequal electron counts, ionic structures, and arbitrary mixtures of selected structures.

5 Direct sampling and structure projection

Introduce the pulled-back factors

$$L_0 = T_\alpha^{\mathsf{T}}L, \qquad F_0 = T_\beta^{\mathsf{T}}F, \tag{25}$$

$$L_H = T_\alpha^{\mathsf{T}}D_\alpha L, \qquad F_H = T_\beta^{\mathsf{T}}D_\beta F, \tag{26}$$

$$L_P = T_\alpha^{\mathsf{T}}E_{\alpha P}L, \qquad G_P = T_\beta^{\mathsf{T}}\sum_Q K_{PQ}E_{\beta Q}F. \tag{27}$$

Combining Eqs. (8) and (24), the needed original product entries are

$$m_{ab} = \sum_{\ell=1}^{r} (L_0)_{a\ell}(F_0)_{b\ell}, \tag{28}$$

$$h_{ab} = \sum_{\ell=1}^{r} \left[(L_H)_{a\ell}(F_0)_{b\ell} + (L_0)_{a\ell}(F_H)_{b\ell} + 2\sum_{P} (L_P)_{a\ell}(G_P)_{b\ell} \right], \qquad (a,b) \in \Omega. \tag{29}$$

For any Davidson input $\mathbf{v}$, build $X = \mathcal{W}\mathbf{v}$ and return

$$(M\mathbf{v})_I = \sum_{(a,b)\in\Omega} w_{ab,I} m_{ab}, \qquad (H\mathbf{v})_I = \sum_{(a,b)\in\Omega} w_{ab,I} h_{ab}. \tag{30}$$

Proposition 2 (Exact factored selected-space action)**.** *Equations* (15)–(30) *evaluate Eq.* (9) *exactly for arbitrary selected structures, without constructing a complete transformed product array.*

Proof. Equation (15) is the exact decomposition into nonzero original columns. Linearity gives Eq. (16), and expansion of the Hamiltonian gives Eq. (24). Applying $\mathcal{T}^*$ to each outer product yields Eqs. (28) and (29). All other final entries have zero coefficient in $\mathcal{W}^*$ by Eq. (5). The final sampling therefore omits no contribution to Eq. (9). □

The contraction can be read as the following sequence:

$$\mathbf{v} \xrightarrow{\mathcal{W}} X = AB^{\mathsf{T}} \xrightarrow{T_\alpha, T_\beta} LF^{\mathsf{T}} \xrightarrow{\text{spin generators}} \text{factor images} \xrightarrow{T_\alpha^{\mathsf{T}}, T_\beta^{\mathsf{T}}} (h,m)|_\Omega \xrightarrow{\mathcal{W}^*} (H\mathbf{v}, M\mathbf{v}). \tag{31}$$

The middle objects are represented by factors and channel panels, not by a selected-space approximation to the transformed wave function. This is how Slater–Condon evaluation remains available without requiring selected structures to form an invariant space.

6 Two additional exact reductions

6.1 Common spin-exchange parity

An optional reduction is available when the two one-spin bases and their orbital maps coincide, and every selected structure has the same certified transpose parity $\eta = \pm 1$. Define

$$\Pi X = X^{\mathsf{T}}, \qquad P_\eta = \tfrac{1}{2}(\mathcal{I} + \eta\Pi), \qquad \Pi\mathcal{W} = \eta\mathcal{W}. \tag{32}$$

Construct X_{rep} by keeping $a < b$ with coefficient $2X_{ab}$, keeping diagonal entries with coefficient X_{aa} for $\eta = +1$, and setting the other entries to zero. Odd parity has zero diagonal. Then

$$P_\eta X_{\text{rep}} = X. \tag{33}$$

Since $\mathcal{T}$ and a spin-independent $\mathcal{H}$ commute with exchange, their original-basis composite $\mathcal{A} = \mathcal{T}^*\mathcal{H}\mathcal{T}$ also commutes. Furthermore $\mathcal{W}^* P_\eta = \mathcal{W}^*$. It follows that

$$\mathcal{W}^*\mathcal{A}X_{\text{rep}} = \mathcal{W}^* P_\eta \mathcal{A} X_{\text{rep}} = \mathcal{W}^*\mathcal{A}P_\eta X_{\text{rep}} = \mathcal{W}^*\mathcal{A}X. \tag{34}$$

The same proof applies to $\mathcal{T}^*\mathcal{T}$ for the overlap. Source factorization may therefore use X_{rep}, but the final output sampling still uses *both* halves of Ω. The unprojected image of X_{rep} is not itself a parity eigenvector.

For a single structure with m independent covalent pairs, this can reduce the source count from 2^m to 2^{m-1}. It does not imply that the touched column count always halves for an arbitrary selected space. Equal spin dimensions alone are insufficient; common parity must be checked from the structure map. Parity is not a substitute for total-spin adaptation. Without it, the unreduced factorization remains exact.

6.2 Causal rows of the exterior adjoint

Only A_Ω rows of the α pullbacks and J rows of the β pullbacks enter Eq. (28). An elementary forward exterior shear updates $z_j \leftarrow z_j + s\epsilon z_i$, with orbital coefficient s and fermionic sign $\epsilon = \pm 1$. Its adjoint updates $z_i \leftarrow z_i + s\epsilon z_j$. Starting from the requested final rows, traverse the adjoint operations in reverse chronological order. Retain an update exactly when its destination is needed, and then mark its source as needed. Execute the retained updates in their original adjoint order, including the diagonal exterior scaling.

For a row selector Q_A, this computes $Q_A T_\sigma^\mathsf{T} Z$ exactly. Every intermediate row that can influence the answer is retained. The pruning concerns paths through a linear transformation, not closure of the selected VB space and not the magnitude of an amplitude. A bitmap on one-spin rows and retained shear-edge indices are sufficient; no product array or dense compound matrix is required.

7 Tiling, storage, and arithmetic

7.1 Streaming the exact equations

Split the r source columns into panels of width s, and the d channels into panels of width p. Different source columns are summed independently in Eq. (28); the Hamiltonian action never couples source column labels to one another. For each source panel:

1. Build its sparse A, B, and apply T_α, T_β to obtain L, F.
2. For a channel panel $\mathcal{P}$, scatter $E_{\sigma P}$ on the corresponding factors.
3. Stream channel panels $\mathcal{Q}$ to accumulate $V_{\sigma P} = \sum_Q K_{PQ} E_{\sigma Q} F_\sigma$, where $F_\alpha = L$ and $F_\beta = F$. This is a channel GEMM on flattened spin-row/source-column panels.
4. Gather $\sum_{P\in\mathcal{P}} E_{\sigma P} V_{\sigma P}$ into the same-spin images. Pull back $E_{\alpha P} L$ and $V_{\beta P}$ and add the cross-spin dot products on Ω.
5. After all channel panels, pull back the base and same-spin factors, add their samples, and reuse the slabs for the next source panel.

Finally apply $\mathcal{W}^*$ once to the accumulated support arrays. Source, channel, and right-hand-side loops reuse shared backing allocations; threads do not receive private copies of the carrier.

The explicit source/channel numerical slabs require

$$O\big((U_\alpha + U_\beta) s p\big) \tag{35}$$

storage, plus $O(t)$ support coefficients and images per processed vector. Additional persistent storage includes the sparse structure map, one-spin link/exterior metadata, and the $d \times d$ kernel K. The source slabs do not include any $U_\alpha U_\beta$ coefficient array. This is a workspace statement, not a bound on the total process memory. For an overlap-only action no channel panels or same-spin Hamiltonian images are required, and the factor workspace reduces to $O((U_\alpha + U_\beta)s)$.

7.2 Where a speed advantage is obtained

Let q_σ count elementary exterior shear edges. With a full channel panel ($p = d$), a leading-work bound per input vector is

$$O\Big(r(U_\alpha + U_\beta)d^2 + rd(q_\alpha + q_\beta) + trd \\ + r\sum_\sigma U_\sigma n_\sigma (n - n_\sigma + 1) + \mathrm{nnz}(\mathcal{W})\Big). \tag{36}$$

The terms describe channel mixing, exterior pullbacks, final sampling, Slater–Condon spin links, and the initial/final structure contractions. The displayed shear count is an upper bound; causal pruning reduces

the selected-row pullbacks. Forward base-factor transforms remain necessary. For $p < d$, channel mixing keeps the same leading arithmetic order, while spin scatters can repeat across channel panels by a factor up to $\lceil d/p \rceil$. A smaller workspace does not change any equation or variational space.

A generated-product contraction instead has channel-mixing work of order $N_{\mathrm{cl}} d^2$, where N_{cl} counts the generated product intermediates needed by that contraction order. When this support is broad and $r \ll \min(U_\alpha, U_\beta)$, the factorized order replaces the product dimension by a source-weighted sum of one-spin dimensions. For $U_\alpha \simeq U_\beta = U$ and $N_{\mathrm{cl}} \simeq U^2$, the leading mixing ratio is approximately $U/(2r)$, before other work is counted.

For broad source support, r can approach U and the factorized order need not be faster. Both orders evaluate the same orthogonal-CI action. An arithmetic planner may select between them without a memory-based admission test for the physical selected space. Exact applicability to arbitrary structures therefore does not mean that their costs are equal or that selecting one structure universally yields rank one. There is no claim of eliminating the exponential growth of the complete one-spin spaces.

8 Correspondence with the current implementation

The equations were checked against revision `a5004add`. Paths below are relative to `src/`; this table records the implementation boundary rather than introducing additional physical assumptions.

Mathematical operation	Implementation
Original support and source columns	`vbscf/structures/assembly/sources.hpp`
Source panels, spin scatter/mix/gather, samples and $\mathcal{W}^*$	`vbscf/structures/assembly/source_tiles.cpp` `apply_source_tiles()`
Hamiltonian channel kernel, Eq. (20)	`mcscf/ci/sigma.cpp` `build_hamiltonian_pair_kernel()`
Slater–Condon spin connections	`mcscf/ci/spin_channels.cpp`
Exterior transforms and causal adjoint rows	`vbscf/structures/orthogonal_ci/exterior_transform.cpp` `projection()` and `apply_adjoint_left_block()`
Independent complete-carrier comparisons	`../tests/vbscf/unit/test_structure_tiles.cpp`

The current change covers fixed-point structure Hamiltonian and overlap actions. Integral adjoints, orbital gradients, and Hessian–vector products have their own existing consumers; the source-factor cost bound is not yet a claim about those complete derivative pipelines. Extending the equations to those consumers requires differentiating the exterior maps, factors, and Hamiltonian consistently, not simply reusing a fixed-point formula.

Main result. Arbitrary selected VB structures can use the orthogonal spin-string Slater–Condon action while Davidson remains in the selected structure space. The exact restriction in Eq. (9) requires no selected-space closure. Equations (24)–(30) make that restriction computable through two-sided spin factors rather than a complete transformed product array. No intermediate projection, coefficient cutoff, or approximate rank compression enters the derivation.